

BUILD WITH AI · CODE FOR COMMUNITIES — SECOND EDITION

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CIVICPULSE BRICS

Scalable AI-Driven Infrastructure Prioritization for Digital Public Goods

“From Citizen Voices to Actionable Infrastructure Intelligence.”

TRACK: OPEN INNOVATION

Team: [Team Name] · Members: [Names + Roles]

CITIZENS -> AI -> SECTOR HEALTH MATRIX
& TELEMETRY -> POLICYMAKERS

Fragmented citizen voices delay action

TODAY

- Voice, text, and messaging feedback is scattered across channels.
- Reports arrive in different languages and regional dialects.
- Unstructured reports are difficult to analyze and deduplicate at scale.
- Critical infrastructure hotspots stay hard to detect in real time.

1. Citizen Reports
2. Data Silos
3. Limited Visibility
4. Delayed Prioritization

CORE GAP

Citizen feedback exists.

Actionable intelligence doesn't.

THE SOLUTION

From feedback to action

CivicPulse BRICS turns citizen reports into geospatially prioritized infrastructure intelligence.

REPORT

Voice / Text / Custom Location & Auto-GPS

UNDERSTAND

AI Transcription + Multilingual Translation

CLUSTER

Geospatial Proximity Hotspots (10km DBSCAN)

PRIORITIZE

Multi-Factor ML Priority Index (0–100 Scorer)

ACT

AI Project Directives, Budgets & Policy Review

Don't just collect complaints. Identify emerging infrastructure needs.

THE PLATFORM

One platform. Five capabilities.

01 Citizen Portal

Voice + Text + Custom Location

Interactive GPS detection & BRICS country selector

02 Multilingual AI

Transcription + Translation

Automatic dialect synthesis & hazard classification

03 Geo-Intelligence

10km Proximity Clustering

DBSCAN grouping of isolated infrastructure requests

04 Priority Engine

Volume + Outage Duration + Sector Deficit + Disruption Intensity

Calculates unified 0–100 ML severity ranking across sectors

05 Policy Command Center

Sector Health Matrix + Autonomous Radar + AI Directives

Interactive radial dials, live threat sweep radar & policy review briefs

TECHNICAL CORE

The CivicPulse intelligence pipeline

01

VOICE /
TEXT

02

MULTILINGUAL
NLU

03

0-100 ML
SEVERITY

04

10KM
DBSCAN

05

PRIORITY
INDEX

06

AI POLICY
BRIEF

Transcribe · Translate · Categorize · 0-100 Multi-Factor Severity Scorer · Geospatial Clustering

TECHNICAL CORE

Multilingual Transformer NLU + Scikit-Learn DBSCAN + Haversine Matrix + Zero-Setup SQLite / PostgreSQL

From data to decision

SECTOR MATRIX & RADAR

Real-time radial gauges for Water, Roads, Power, Sanitation & autonomous threat radar sweep.

LIVE FEED & TELEMETRY

Anonymized, AI-translated citizen requests with live institutional status tickers.

AI POLICY DIRECTIVES

Autonomous project solutions with actionable budgets, timelines, and policy review citations.

1

50 Citizen Reports

2

1 Geographic Hotspot

3

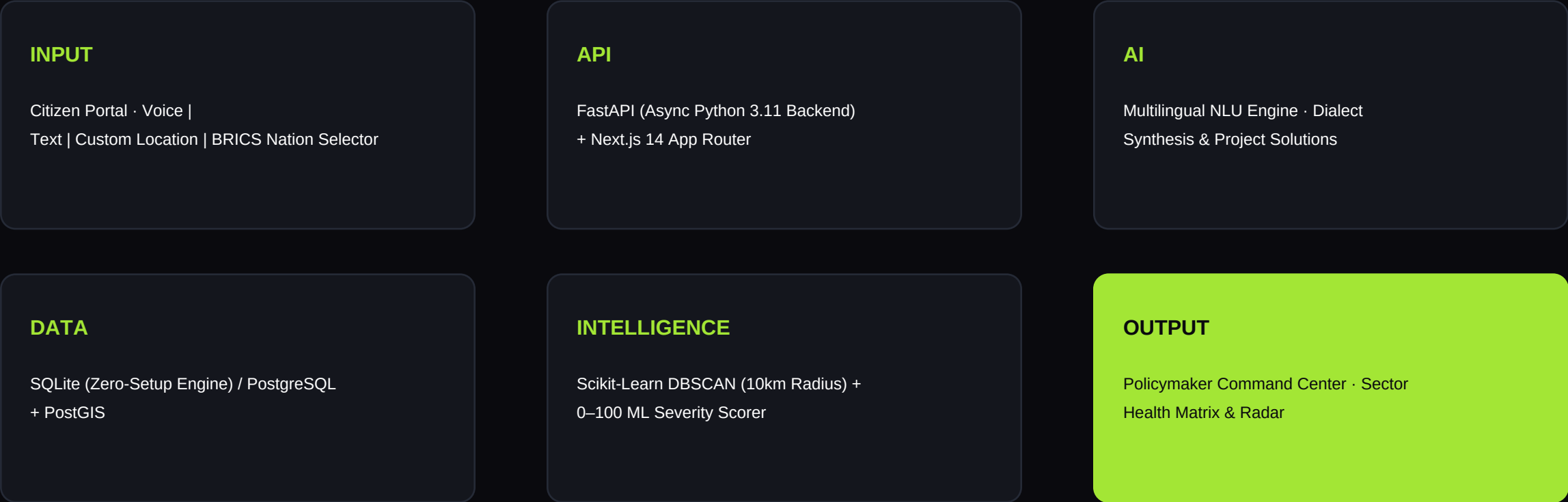
1 Priority Project

4

Policy Action

SYSTEM DESIGN

Simple, scalable architecture



PRODUCTION-READY BY DESIGN

Built for real-world civic systems

PRIVACY

PII sanitization & anonymization
before persistence

SECURITY

JWT + HTTP-only cookies
& input schema validation

ACCESS CONTROL

RBAC: SuperAdmin | Policymaker
| Analyst | Citizen

SCALABILITY

Local ward -> Regional
-> National -> BRICS-scale

SUCCESS METRICS

Processing Time · Cluster Quality · Multi-Factor ML Accuracy · PII Removal Effectiveness

THE END-TO-END PROMISE

One report -> one actionable decision

1. Citizen Voice & Text Report
2. AI Multilingual Audio Synthesis
3. Automatic Translation & Hazard Tagging
4. PII Sanitization & Anonymization
5. Geospatial Clustering (10km DBSCAN)
6. 0–100 Multi-Factor ML Severity Scoring
7. AI Project Recommendation with Budget
8. Sector Health Matrix & Policy Action

REAL-WORLD EXAMPLE

“Pipeline rupture reported in Ward 4”

-> Hotspot detected (10km DBSCAN)

-> ML Severity 88/100 (Critical Hazard)

-> Project Brief auto-drafted with \$85,000 budget & 4-day timeline

CIVICPULSE BRICS

“Turning Citizen Voices into Actionable Infrastructure Intelligence.”

Build with AI: Code for Communities — Second Edition

Multilingual AI · Geospatial Intelligence · Digital Public Goods